Relatorio de Segmentacao de Poros

EDS / CBS / QEMSCAN - Analise Morfometrica PARTISAN

Imagem analisada : S1.png (Grão 1)

Data de geracao : 30/03/2026 18:57

Porosidade total : 45.118%

Poros analisados : 76

Gerado por: pore_segmentation (Ilastik 1.4 + PARTISAN v2.0)

Parametros de Configuracao

```
ilastik.ilastik_exe: D:\WILLIAM\ilastik-1.4.1.post1\ilastik.exe
ilastik.project_path: D:\WILLIAM\PHYGEO2026\MORPHO\pore_segmentation\MyProject.ilp
ilastik.n_threads: 4
ilastik.ram_mb: 4096
ilastik.pore_class_index: 0
preprocessing.tile_size: 0
preprocessing.normalization: percentile
preprocessing.norm_percentile_low: 1
preprocessing.norm_percentile_high: 99
postprocessing.threshold_method: otsu
postprocessing.threshold_value: 0.5
postprocessing.morphology_opening_radius: 2
postprocessing.morphology_closing_radius: 2
postprocessing.morphology_fill_holes: True
postprocessing.min_pore_area_px: 50
postprocessing.max_pore_area_px: 0
output.output_dir: results
output.export_formats: ['csv', 'excel']
output.generate_overlay: True
output.generate_report: True
output.overlay_colormap: jet
output.figure_dpi: 150
logging.level: INFO
```

Resumo da Segmentacao

Total de poros detectados : 76

Porosidade total : 45.1179%

Area media dos poros : 5693.1 px

Area mediana dos poros : 579.0 px

Maior poro : 68345 px

Menor poro : 51 px

Metricas PARTISAN - medias:

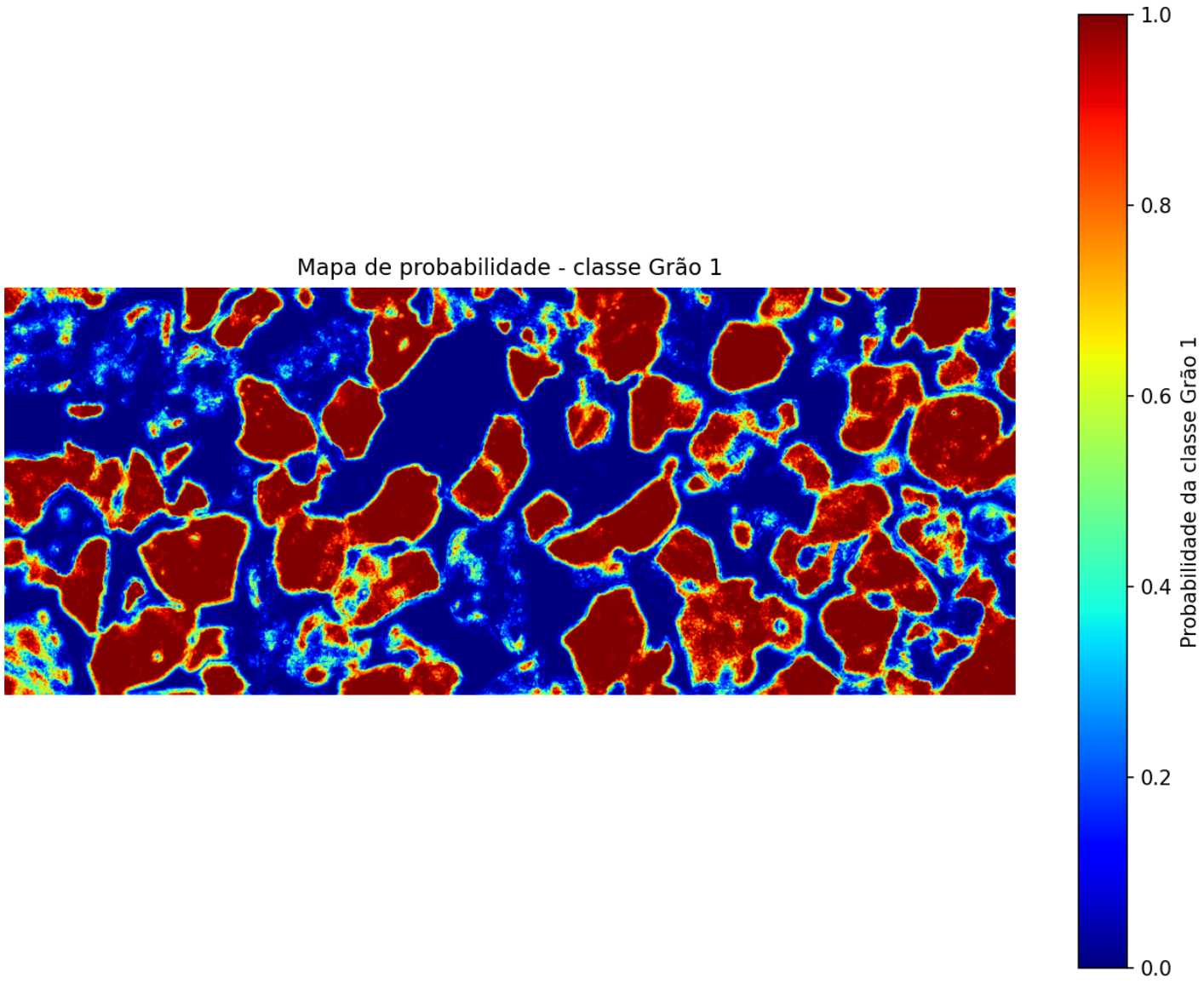
CI_Circ (circularidade) : 0.7540

CI_AR (razao aspecto) : 1.9345

CI_Sol (solidez) : 0.8415

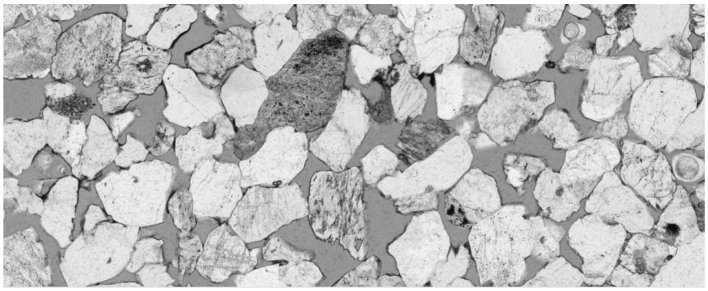
FF (form factor) : 0.7540

Probabilidade

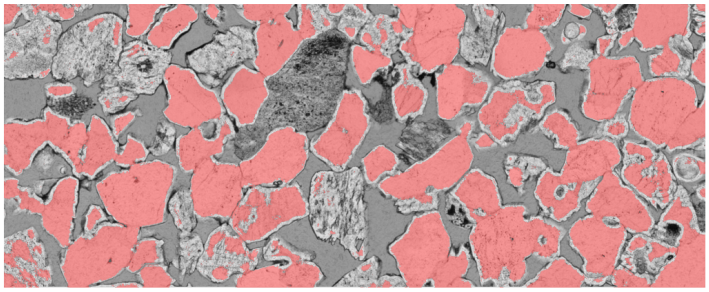


Overlay

Imagem original

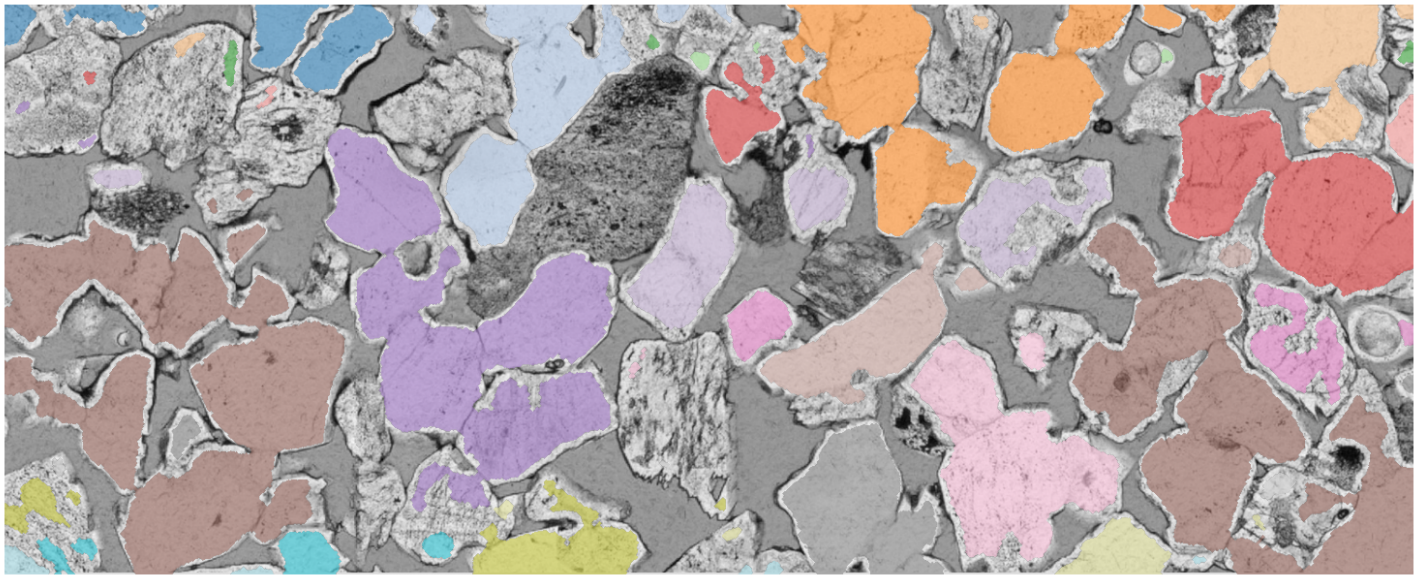


Classe Grão 1 segmentada - fracao: 45.12%

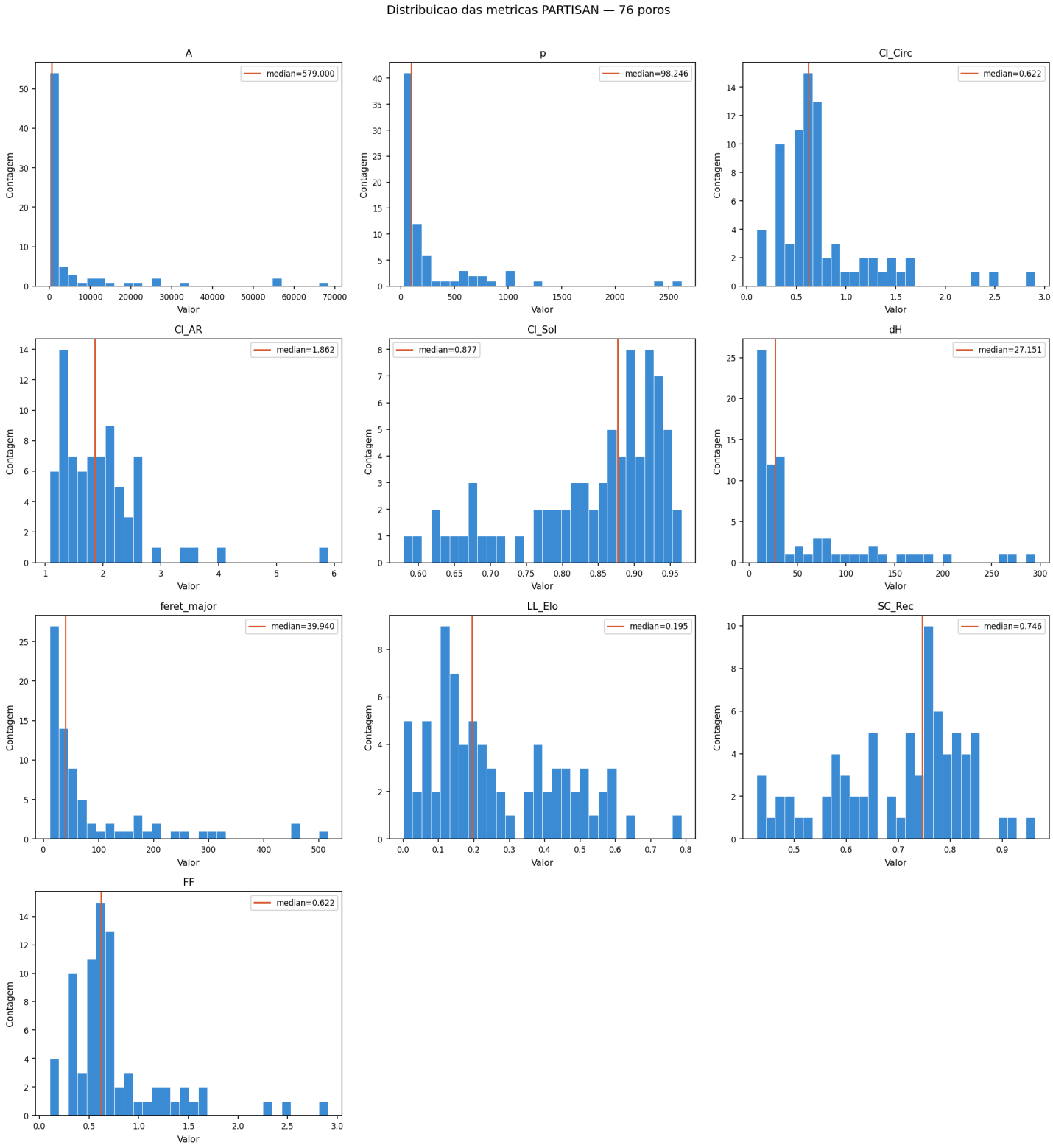


Labels

Poros individuais (N=83)



Histogramas



Estatisticas Descritivas - Metricas PARTISAN

Metrica	Count	Mean	Std	Median	Min	Max
p	76	282.6160	466.9113	98.2462	20.9706	2620.4604
A	76	5683.1184	13002.8923	579.0000	51.0000	68345.0000
w	76	51.0447	69.2748	21.1066	5.0000	376.0000
b	76	81.9228	105.1238	37.4454	8.4853	511.7254
c	76	173.2673	204.8085	85.2975	25.3157	926.7408
a	76	71.8816	81.0499	36.5000	10.0000	347.0000
m	76	32.1807	40.0688	15.2536	3.9231	235.4643
Lb	76	81.8816	105.9021	36.0000	9.0000	516.0000
Wb	76	59.5658	78.4662	23.5000	6.0000	349.0000
dH	76	55.1527	65.1926	27.1510	8.0582	294.9908
pHull	76	199.2600	250.9058	87.8614	20.1421	1263.1585
Ahull	76	8162.8289	20362.6341	646.5000	57.0000	109634.0000
ece	76	241.7716	308.6554	107.4875	27.6335	1355.9026
L_maj	76	85.0956	108.5374	38.4400	10.8297	499.7622
L_min	76	47.0301	61.9532	20.6090	5.7114	329.5549
d_BC	76	86.5106	109.5412	38.9404	10.1980	516.0087
feret_major	76	87.0801	109.0400	39.9402	11.1980	517.0080
feret_minor	76	51.0447	69.2748	21.1066	5.0000	376.0000
w_f	76	81.9228	105.1238	37.4454	8.4853	511.7254
DL_Circ	76	1.3195	0.4364	1.2678	0.5865	3.1302
DL_Rec	76	0.9194	0.2353	0.9503	0.3979	1.7881
DL_Com	76	0.7038	0.1266	0.7459	0.4283	0.9630
DL_Elo	76	2.4760	1.1043	2.3351	1.0363	8.3478
CI_Circ	76	0.7540	0.5069	0.6221	0.1021	2.9067
CI_AR	76	1.9345	0.7425	1.8623	1.0758	5.8927
CI_Con	76	1.0780	0.3734	0.9424	0.5110	2.5512
CI_Sol	76	0.8415	0.1012	0.8765	0.5788	0.9657
LL_Circ	76	0.8303	0.2558	0.7887	0.3195	1.7049
LL_AR	76	0.7380	0.1878	0.8048	0.2128	1.0000
LL_Elo	76	0.2620	0.1878	0.1952	0.0000	0.7872
LL_Con	76	0.8760	0.1232	0.9230	0.3996	1.0000
LL_Sol	76	0.8415	0.1012	0.8765	0.5788	0.9657
LI_AR	76	0.5727	0.1670	0.5370	0.1697	0.9296
LI_Con	76	0.8760	0.1232	0.9230	0.3996	1.0000
LI_Sol	76	0.8415	0.1012	0.8765	0.5788	0.9657
SC_Circ	76	0.4894	0.1471	0.4889	0.1591	0.7937
SC_Rec	76	0.7038	0.1266	0.7459	0.4283	0.9630
SC_ARF	76	0.6353	0.1920	0.6096	0.1915	1.0000
SC_AR	76	1.7501	0.6642	1.6407	1.0000	5.2222
FF	76	0.7540	0.5069	0.6221	0.1021	2.9067
Reg	76	0.2839	0.2314	0.2497	0.0137	1.5098